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# NAIVE BAYES

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
import pandas as pd

iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)
df["Species"] = iris.target

x = df.drop(columns=["Species"])
y = df["Species"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)

model = GaussianNB()
model.fit(x_train, y_train)

# Prediction
y_pred = model.predict(x_test)
print("Predictions:", y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred) * 100)

print("Confusion Matrix")
print(confusion_matrix(y_test, y_pred))

Predictions: [2 2 1 2 2 2 1 1 1 1 1 1 1 1 2 2 0 1 2 1 0 0 0 0 2 2 0 0 1 2 2]
Accuracy: 90.0
Confusion Matrix
[[ 7  0  0]
 [ 0 10  2]
 [ 0  1 10]]
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# SVM

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix
import pandas as pd

iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)
df["Species"] = iris.target

x = df.drop(columns=["Species"])
y = df["Species"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)

model = SVC(kernel="linear", C=1.0)
model.fit(x_train, y_train)

y_pred = model.predict(x_test)
print("Predictions:", y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred) * 100)

print("Confusion Matrix")
print(confusion_matrix(y_test, y_pred))

Predictions: [2 0 0 0 1 2 1 0 1 2 0 1 1 0 2 1 2 1 2 2 0 0 0 2 2 0 1 0 0 0]
Accuracy: 93.33333333333333
Confusion Matrix
[[13  0  0]
 [ 0  6  0]
 [ 0  2  9]]

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